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Improving the classification accuracy for IR spectroscopic diagnosis of stomach and colon malignancy using non-linear spectral feature extraction methods†

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Non-linear feature extraction methods, neighborhood preserving embedding (NPE) and supervised NPE (SNPE), were employed to effectively represent the IR spectral features of stomach and colon biopsy tissues for classification, and improve the classification accuracy for diagnosis of malignancy. The motivation was to utilize the NPE and SNPE's capability of capturing non-linear spectral behaviors by simultaneously preserving local relationships in order that minute spectral differences among classes would be effectively recognized. NPE and SNPE derive an optimal embedding feature such that the local neighborhood structure can be preserved in reduced spaces (variables). The IR spectra collected from stomach and colon tissues were represented by several new variables through NPE and SNPE, and also by using the principal component analysis (PCA). Then, the feature-extracted variables were subsequently classified into normal, adenoma and cancer tissues by using both *k*-nearest neighbor (*k*-NN) and support vector machine (SVM), and the resulting accuracies were compared with each other. In both cases, the combination of SNPE-SVM provided the best classification performance, and the accuracy was substantially improved compared to when PCA-SVM was used. Overall results demonstrate that NPE and SNPE could be potential feature-representation strategies useful in biomedical diagnosis based on vibrational spectroscopy where effective recognition of minute spectral differences is critical.

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Introduction

Vibrational spectroscopic methods such as infrared (IR) and Raman microscopy have been studied extensively for diagnosing various diseases including cancers by examining spectral features obtained from biopsy tissues.[1–27] These methods have been adopted in biomedical diagnosis to utilize rich chemical, pathological information related to differences between normal and malignant sections, thereby possibly improving the identification accuracy and objectivity of diagnosis particularly when used in combination with the conventional microscopic inspection of a stained biopsy sample. Review papers that summarize the applications of vibrational spectroscopy in the field of biomedical diagnosis are readily available.[3,4,14,15,19]

Since human tissues are composed of many sub-units with widely different molecular structures, the corresponding vibrational spectral features are broad because of the extensive overlapping of individual bands. Thus, the spectral signature indicative of malignancy is indistinct. Therefore, to effectively recognize minute spectral differences between normal and malignant sections, multivariate classification methods such as partial least squares-discriminant analysis (PLS-DA), *k*-nearest neighbor (*k*-NN) and support vector machine (SVM) have been widely adopted.[13,16] To improve the accuracy of diagnosis, enhancing the spectral selectivity (discriminability) among target groups is a straightforward way. However, once a spectroscopic method is chosen for biomedical diagnosis, the increase of spectral resolution is the only alternative approach.

A practical alternative is to use the acquired spectral features effectively to enhance discrimination among classes. In general, the number of variables (dimensions) in vibrational spectra is large, and the multi-collinearity is usually present among the variables. In this situation, if the original spectral characteristics could be effectively translated and represented by using fewer variables without collinearity, the classification analysis

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becomes simpler and more robust. For this purpose, one of the most adopted methods is the principal component analysis (PCA), representing the original spectral data into a few variables (scores) by projecting them into eigenvectors (factors) orthogonally constructed to maximize the variation present in a given spectral dataset.[28] Orthogonality among the constructed factors intrinsically eliminates the problem of collinearity. Due to the above-mentioned merits, PCA has been extensively used as a feature representation method, particularly in the field of vibrational spectroscopic disease diagnosis.[1,5,10,16]

Recently, several new feature extraction methods based on nonlinear dimensionality reduction algorithms have been developed, such as locally linear embedding (LLE), Laplacian eigenmaps, local tangent space alignment and conformal eigenmaps, that can deal with nonlinear behavior in a dataset and reveal the hidden structure more effectively.[29–32] These methods are capable of capturing the nonlinear structure by preserving the local relationships between data points, thus improving the performance of feature extraction in the presence of outliers. However, it is unable to generate a loading matrix that is essential to build a prediction model since the coordinates mapping in these methods is defined only for training samples, but not for testing (unknown) samples. This is a severe drawback for a prediction method to be used for actual diagnosis of new biopsy samples. To resolve this problem, the neighborhood preserving embedding (NPE) has been proposed as a linear approximation to LLE algorithm which can preserve the local neighborhood structure.[33] As shown NPE can be less sensitive to the presence of outliers than PCA.[33] Although the non-linear feature extraction methods such as NPE can be more versatile in representing detailed spectral variations, they have not been readily exploited in the field of disease diagnosis based on vibrational spectroscopy.

For diagnosis of stomach and colon malignancy (adenoma/cancer), we evaluated NPE as a method of extracting characteristic features from the IR spectra of corresponding biopsy tissues. The supervised-NPE (SNPE) method was also evaluated to determine whether it is more effective in feature extraction since it uses prior label (class) information in defining the local neighborhood structure. The further mathematical description of NPE and SNPE will follow in the Experimental section.[34] PCA as a linear feature extraction method was also performed for the purpose of comparison. PCA aims to preserve the total variance, while NPE aims to preserve the local neighborhood. Initially, the IR spectra collected from both stomach and colon tissues were represented by new variables (coordinates) using PCA, NPE and SNPE. Then, in each case, the feature-extracted variables were subsequently used to classify normal, adenoma and cancer tissues using both *k*-NN and SVM, and the resulting accuracies were evaluated and compared with each other. *k*-NN is chosen because it is among the simplest classification methods, and the performance of SVM as a classifier has been well-proven in many domains. SVM is a popular and powerful supervised learning method for classification, which finds the best hyperplane to classify samples by maximizing the distance between class boundaries. Basic SVM provides a discriminant function for two-group classification and a nonlinear class boundary can be obtained when a nonlinear kernel function is employed. For multi-class classification, the "one against the rest" method that extends binary classifiers is mostly adopted.[36] As expected, the different characteristics of each method for feature extraction should directly influence the resulting accuracy of classification. Finally, we compared the variables represented by each method to describe the differences in the corresponding classification performances.

Experimental

IR spectral collection and analysis of spectral datasets

For the stomach dataset, the IR spectra of the stomach tissues collected in a previous publication were used.[18] As shown in Table 1, it consists of a total of 926 spectra including normal, adenoma and cancer sections acquired from eleven different patients. The biopsy samples were obtained from eight male and three female subjects, and the ages of subjects ranged from 47 to 65 years old.

For the preparation of colon dataset, the normal and malignant (adenoma/cancer) tissues from three male patients (ages from 51 to 62 years old) were obtained from Guro Medical Center of Korea University, Seoul, Korea. The IR spectra (resolution: 4 cm^{-1}, 32 scans) were collected using an FT-IR spectrometer equipped with a microscope that has an aperture size of 25 × 25 μm (SensIR Technologies, Danbury, CT). As performed in the previous study,[18] the samples were cut (10 μm in thickness) and transferred onto an IR-reflective e-glass for measurement. Two consecutive sections were obtained from each biopsy sample. Using the first section, the target spots for IR spectral collection were determined based on diagnostic information from the pathology testing (H&E stained sample). After the target spots were established, then actual IR spectra were collected for the second (a parallel unstained) section. Even though these two samples were sectioned consecutively,

Table 1 The assigned number of IR spectra into the training and validation sets for stomach and colon datasets. The numbers in the parentheses indicate the selected subjects

		Number of spectra in the training set	Number of spectra in the validation set
Stomach	Normal	221 (subject # 1, 3, 8, 9, 10)	127 (subject # 2, 5, 7)
	Adenoma	147 (subject # 1, 9, 10)	29 (subject # 2)
	Cancer	297 (subject # 3, 4, 6, 8, 11)	105 (subject # 5, 7)
Colon	Normal	222 (subject # 2, 3)	62 (subject # 1)
	Adenoma	79 (subject # 2, 3)	65 (subject # 1)
	Cancer	130 (subject # 2, 3)	45 (subject # 1)

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the pathological presentations of them could be slightly different from each other. Hence, to secure the correct pathological identity of the measured spots, the second section was then stained after the spectral collection and the measured spots were pathologically re-examined.

The division of datasets into the training and validation sets is described in Table 1. In the case of the stomach dataset, it was intended to assign IR spectra to have a similar percentage in the training and validation sets for all three cases (normal, adenoma and cancer). All of the spectra acquired from a patient were assigned only to either the training or validation set for both cases.

In the case of the colon dataset, the IR spectra acquired from two subjects were assigned to the training set, and the other spectra acquired from the one remaining subject were served as the validation set. The numbers in the parentheses indicate the selected subjects for the division of datasets. All the spectral pretreatments (baseline correction and area normalization), feature extractions (PCA, NPE and SNPE) and discrimination analyses (*k*-NN and SVM) were performed using MATLAB R2007a (Math Works Inc., MA, USA). The value of *k* in *k*-NN and the number of factors in PCA were determined by cross-validation on the training set. Gaussian kernel was used in SVM, and the bandwidth and other parameters were also determined by the cross-validation.

Brief mathematical description of NPE and SNPE

NPE is one of the nonlinear feature extraction methods for dimensionality reduction. The reduction of data dimensions is frequently employed when data composed of large variables such as spectroscopic data are analyzed, because it helps to mitigate away from the curse of dimensionality. NPE is a linear approximation to LLE algorithm which can preserve the local neighborhood structure on data manifold, thereby obtaining a loading matrix by introducing the linearity assumption. In NPE, a weight matrix $W = (w_{ij})$ in eqn (1) describing the linear combination coefficients of the nearest neighbor points is initially obtained by minimizing the reconstruction error defined below:

$$\varepsilon = \sum_{i=1}^{N} \left\| x_i - \sum_j w_{ij} x_j \right\|^2 \quad (1)$$

where x_i is the ith data point and represents L^2-norm of a column vector. Let y_i be the data point in the reduced space corresponding to x_i. It is assumed that $y_i = A^{\mathrm{T}} x_i$ for a loading matrix A which is the $D \times d$ matrix, where D and d are the original dimensionality and the reduced dimensionality, respectively. Then, the loading matrix A is derived such that the local neighborhood structure can be preserved in the reduced space through the minimization of the following cost function:

$$\Phi = \sum_{i=1}^{N} \left\| y_i - \sum_{j=1}^{N} w_{ij} y_j \right\|^2 = \sum_{i=1}^{N} \left\| A^T x_i - \sum_{j=1}^{N} w_{ij} A^T x_j \right\|^2 \quad (2)$$

under the constraint $a^{\mathrm{T}} x x^{\mathrm{T}} a = c$, where a is a column vector of A and c is a positive constant. Then NPE obtains its low dimensional representation y by $y = A^{\mathrm{T}} x$ when new data arrive.[33,35]

A supervised version of NPE,[34] SNPE, utilizes prior label (class) information of dataset in the dimension reduction process, while NPE performs the same reduction process without the label information. So, SNPE finds the nearest neighbors within the same label. Under the assumption that class distributions in a training set represent the original population distributions, SNPE shows a better classification performance. For a more in-depth understanding of NPE and SNPE, the references are of great help.[33,34]

Results and discussion

IR spectral features of colon tissues

Fig. 1 shows the normalized colon IR spectra pertinent to normal, adenoma and cancer sections obtained by averaging all of the corresponding spectra acquired from all three subjects. Before averaging the spectra, the baselines of raw spectra were corrected and zeroed at both 1800 and 900 cm^{-1}, and the baseline-corrected spectra were divided by the corresponding area under the 1800–900 cm^{-1} range. By using the area normalization, the unwanted spectral variations caused by different sample thicknesses at the measurement spots could be reduced as described.[18] The overall spectral features of colon tissues are quite similar to those of stomach tissues shown in the previous study.[18] It indicates the overall constituents consisting of colon and stomach tissues are not significantly different from each other. The absorption features of colon tissues on the right side of the amide II band (centered at 1540 cm^{-1}) as well as in the 1160–900 cm^{-1} range, where the stretching and deformation of glycogen are largely present, are different from those of stomach tissues.[3]

When the IR spectra of normal and malignant (adenoma and cancer) colon tissues are compared, the distinct spectral differences are observed in the 1800–1500 and 1160–900 cm^{-1} ranges. In the 1800–1500 cm^{-1} range, the positions and shapes of amide I (1660 cm^{-1}) and amide II (1540 cm^{-1}), indicative of

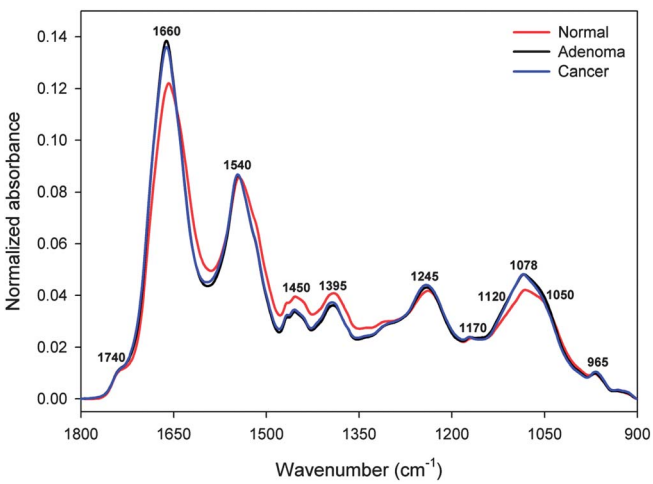

Fig. 1 Normalized colon IR spectra corresponding to the normal (red), adenoma (black) and cancer (blue) sections obtained by averaging all of the corresponding spectra acquired from all three subjects.

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the protein structure as found in the literature,[3] are different from each other. In the case of malignancy, the band position slightly shifts to the higher wavenumber, and the width of both bands also becomes narrower. The varied spectral features of amide I and II bands clearly demonstrate that alteration of the protein structure is evident in the malignant tissues. In the 1160–900 cm^{-1} range, the 1078 cm^{-1} band corresponding to the C–C stretching of glycogen is present.[3] The absorption of this band in normal and malignant tissues greatly differs, thus indicating a large variation of glycogen content when a tissue becomes malignant.

Overall, moderate spectral differences between normal and malignant colon tissues are observed. Although the biological origin of spectral differences is difficult to fully address at present, the observed spectral differences are valuable for the identification of colon malignancy. While, the spectral features between adenoma and cancer are nearly analogous as shown in Fig. 1. Therefore, their differentiation could be highly challenging. In contrast, in the case of stomach tissue, relatively more discernible spectral differences between adenoma and cancer were observed as shown in the previous publication.[18]

Comparison of feature-extracted variables by PCA, NPE and SNPE

Before proceeding further to classification, it is necessary to observe new variables constructed using PCA, NPE and SNPE to examine how they differ from one another in the representation of IR spectral features. For this purpose, we compared the 1st and 2nd variables (scores), and the most significant variables, extracted using each method. Fig. 2(a)–(c) show the scatter plots of the 1st *vs.* 2nd scores extracted from the stomach calibration dataset using PCA, NPE and SNPE, respectively. Red, white and blue circles correspond to normal, adenoma and cancer sections, respectively. As shown in the figures, the two-dimensional distributions of scores in each case are quite different from each other. This clearly indicates that the characteristics of these three methods in the spectral feature representation are different.

In the PCA score scatter plots, the scores of normal (red circles), adenoma (white circles) and cancer (blue circles) are located at the center with wide distribution and they largely overlap with each other; while, the scores of adenoma are relatively more localized at the left side. As shown in the figures, the discrimination among the three classes is not distinct in the domain of PCA scores. The most significant observation in the NPE score scatter is that several localized clusters appear, particularly on the left part of the scatter. As the scores are more concentrated in each cluster, the relatively more distinct discrimination among the classes becomes. It is clear that NPE starts to recognize more detailed spectral features, for example, possible subject-to-subject spectral differences. In the SNPE score scatter, the formation of more clusters is observable. Since SNPE is a supervised method utilizing prior class information in determining local neighbors, more localized scores were found as expected.

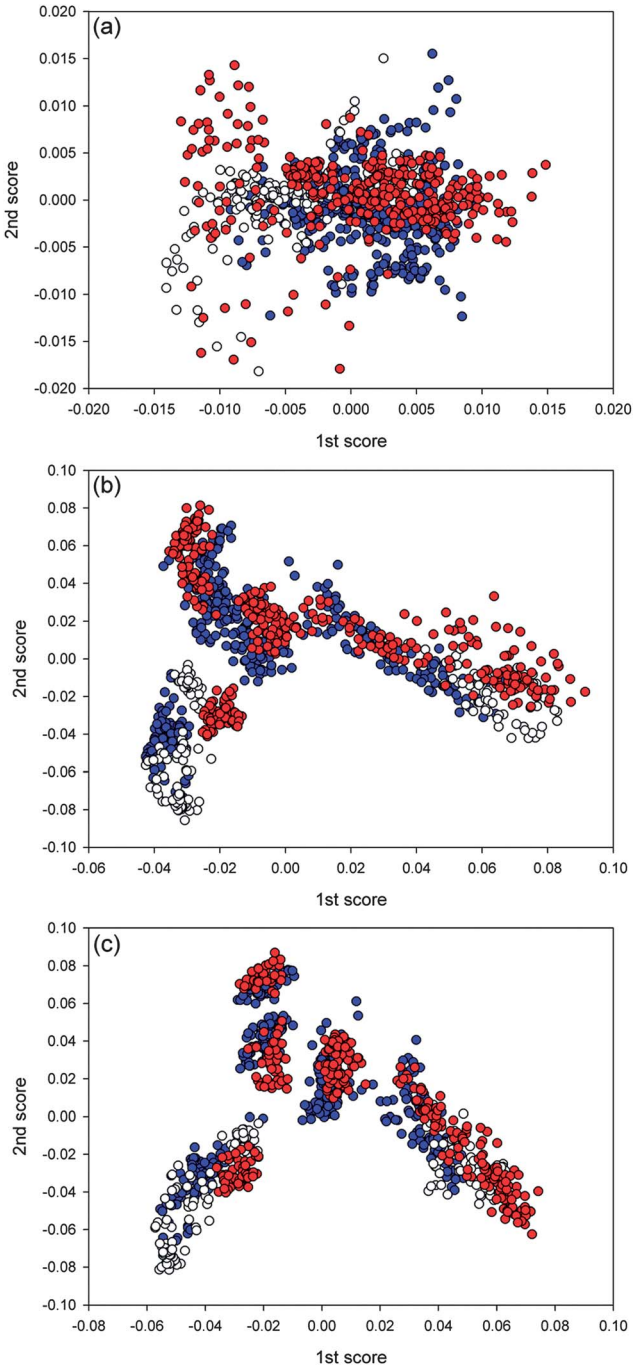

Fig. 2 Scatter plots of the 1st *vs.* 2nd scores extracted from the stomach calibration dataset using PCA, NPE and SNPE, respectively. Red, white and blue circles correspond to normal, adenoma and cancer sections.

Fig. 3 (a)–(c) show the scatter plots of the 1st *vs.* 2nd scores extracted from the colon calibration dataset using PCA, NPE and SNPE, respectively. Red, white and blue circles correspond to normal, adenoma and cancer sections, respectively. As was similarly observed in Fig. 2, the distributions of scores in each case are quite different from each other, and their variations become more concentrated and clustered when both NPE and SNPE are used. As expectable from the spectra in Fig. 1, the scores of adenoma and cancer are closely located and greatly

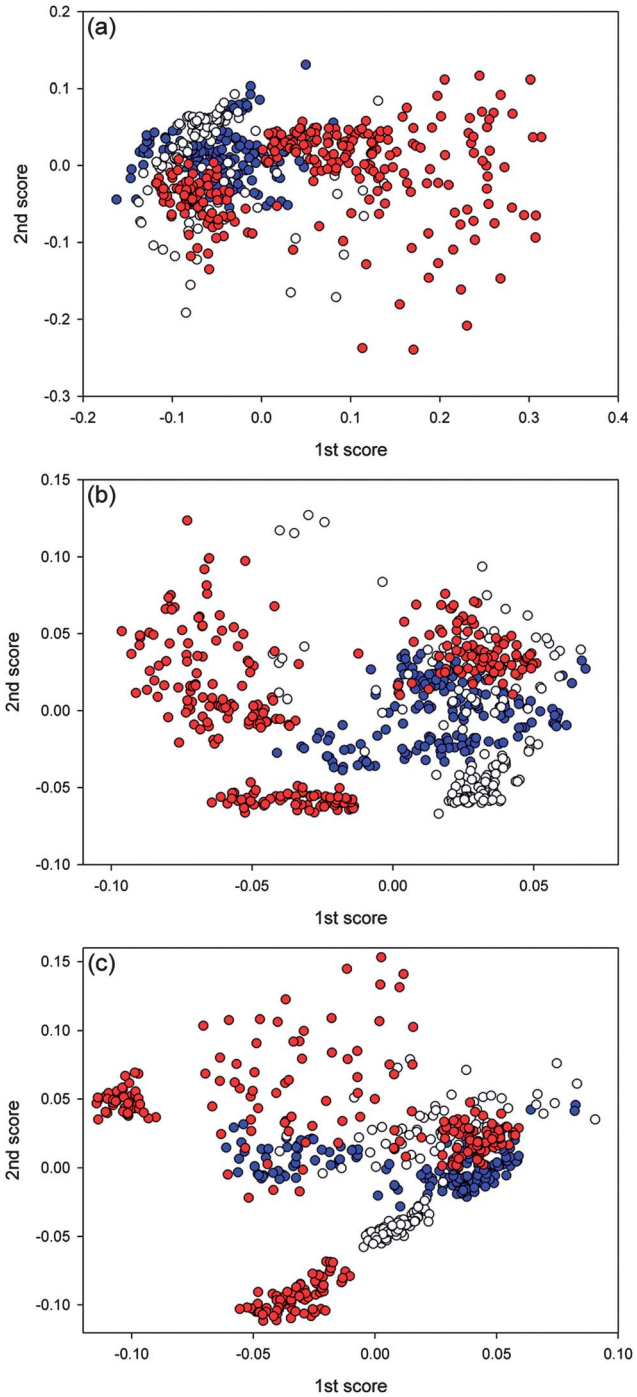

Fig. 3 Scatter plots of the 1st *vs.* 2nd scores extracted from the colon calibration dataset using PCA, NPE and SNPE, respectively. Red, white and blue circles correspond to normal, adenoma and cancer sections.

overlapped, and so their differentiation is not easy. In the meanwhile, the discrimination between normal and malignant colon tissues is relatively feasible.

The observations of Fig. 2 and 3 clearly demonstrate that, in comparison with PCA, both non-linear feature extraction methods recognize the spectral variation differently, and their representation is more reflective to detailed spectral features present in the datasets. When the feature-extracted variables from three methods are used for subsequent classification, the resulting accuracies are expected to be clearly different, and the use of variables describing the detailed spectral variations is advantageous to improve classification accuracy.

Comparison of classification accuracies resulted from the use of PCA, NPE and SNPE

Table 2 shows the accuracies of classifying stomach tissues into normal, adenoma and cancer groups using *k*-NN and SVM, when new feature-extracted variables by PCA, NPE and SNPE are separately employed. The overall accuracy corresponds to the percentage of all correctly predicted (true positive) samples regardless of the classes over the total samples in the validation sets. Individual accuracies in identification of each class are also listed for comparison. For all three feature extraction methods, 483 dimensions (variables) of the original IR spectra were reduced to 10 dimensions, which were found to be an optimum by the cross-validation. In each method, the optimized values of *k* used for *k*-NN, and values of *C* and bandwidth *h* of Gaussian kernel used for SVM are summarized in the ESI.† Also, diagnostic performance measures on the classification in the case of using SVM are shown in the ESI.†

The classification models based on the variables constructed using NPE and SNPE outperformed the PCA-based models. In the use of a given classifier (either *k*-NN or SVM), the accuracies were improved when NPE and SNPE were used, indicating their superior spectral feature representation for the classification. When SNPE and SVM were combined, the classification was most accurate with an accuracy of 88.1%. In general, SNPE is expected to be more descriptive than NPE due to the supervised learning. The classification performance of SVM is generally better than that of *k*-NN, since SVM computes class boundaries by globally recognizing a pattern in a dataset, while *k*-NN considers only a limited area around a data point in an implicit manner for computing physical distance. The connection of both SNPE and SVM advantages contributes to the better performance in classification. Compared to the accuracy achieved by using PCA-SVM (77.8%), the SNPE-SVM classification accuracy is increased by a margin of 10.3%.

Table 3 shows the accuracies in classifying colon tissues into normal, adenoma and cancer groups using *k*-NN and SVM, when new feature-extracted variables by PCA, NPE and SNPE are employed. For all three cases, the use of 5 dimensions was

Table 2 Accuracies of classification using *k*-NN or SVM for normal, adenoma and cancer stomach tissues when three different feature extraction methods of PCA, NPE and SNPE are employed

	Number of dimensions	Classifier used	Overall	Normal	Adenoma	Cancer
PCA	10	*k*-NN	74.0%	53.5%	89.7%	94.3%
	10	SVM	77.8%	66.1%	100%	85.7%
NPE	10	*k*-NN	82.8%	80.3%	89.7%	83.8%
	10	SVM	86.6%	81.1%	75.9%	96.2%
SNPE	10	*k*-NN	77.0%	58.3%	96.6%	94.3%
	10	SVM	88.1%	92.1%	100%	80.0%

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Table 3 Accuracies of classification using *k*-NN or SVM for normal, adenoma and cancer colon tissues when three different feature extraction methods of PCA, NPE and SNPE are employed

	Number of dimensions	Classifier used	Overall	Normal	Adenoma	Cancer
PCA	5	*k*-NN	39.0%	75.8%	3.1%	40.0%
	5	SVM	47.1%	85.5%	3.1%	57.8%
NPE	5	*k*-NN	61.6%	100%	0.0%	97.8%
	5	SVM	59.3%	100%	0.0%	88.9%
SNPE	5	*k*-NN	54.1%	77.4%	6.2%	91.1%
	5	SVM	61.6%	100%	0.0%	97.8%

optimal. Optimized values of k, C and h, and diagnostic performance measures on the classification in the case of using SVM are also shown in the ESI.† The use of fewer dimensions in the representation of colon spectra is reasonable since the training dataset consists of spectra collected from only two subjects. Similarly, the classification accuracy is improved when the non-linear feature extraction methods are used. The best classification performance is achieved by using either NPE-*k*-NN or SNPE-SVM. The resulting accuracy is 61.6%, which means an absolute increase of 14.5% compared to that achieved by using PCA-SVM (47.1%). As shown in Table 3, the discrimination of colon adenoma from cancer was unsuccessful. The malignant sections were mostly classified into the cancer group, consequently degrading the overall classification accuracy in the diagnosis of colon malignancy.

In addition to *k*-NN and SVM, the performance of linear discriminant analysis (LDA), another popular classifier, was also tested. When LDA was applied to classify stomach tissue samples, the resulting accuracies using PCA, NPE and SNPE scores were 63.6, 64.8 and 67.8% respectively. In the case of colon tissues, the accuracies were 45.9, 47.6 and 49.4% when PCA, NPE and SNPE scores were used for the classification, respectively. Comparing with *k*-NN and SVM, the inferior performance of LDA is due to the fact that the scores of the same class are fairly spread out in the domain as shown in Fig. 2 and 3, so it is inefficient to differentiate the classes by linear class-boundary.

Conclusions

Nonlinear feature extraction methods of NPE and SNPE effectively recognized the minute IR spectral differences between normal and malignant sections. When NPE and SNPE were harmonized with SVM, the classification accuracy for the diagnosis of stomach/colon malignancy improved compared with the results using the PCA-based linear extraction method. The description and representation of vibrational spectral features can be more effective when nonlinear feature extraction methods are used rather than the linear-based methods.

Although the results shown here are not sufficient to generalize the superiority of NPE and SNPE in the feature extraction, we suggest that NPE and SNPE are potential feature-representation strategies that could be quite useful in biomedical diagnosis based on vibrational spectroscopic methods such as IR and Raman spectroscopy. Our future research will be directed to evaluate these methods to enhance the contrast of vibrational spectroscopic images acquired from biopsy tissues for more accurate diagnosis.

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